Sanjana Shah

Ahmedabad, India | Open to relocation | Immediate Joiner
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SUMMARY

Software engineer with 2+ YOE at Meta and three summer internships at Amazon, building AWS backend services, mobile networking infrastructure, and full-stack tools using C++, Java, and Python. Seeking backend and infra engineering roles in India.

EXPERIENCE

Meta Mar 2024 – Aug 2026
Software Engineer New York, NY, USA

- Cut global load-balancing convergence time by **15 minutes** and reduced PoP CPU overprovisioning by **5%**, saving an estimated **$2–3M annually**, by building and rolling out a C++ DNS cache-refresh timer for Instagram and Threads on iOS and Android, validating disaster-recovery behavior through Python telemetry and A/B testing.
- Reduced compressed binary size by **438–472 KiB per app** across three production iOS apps by analyzing dependencies with Buck, validating dead code through breakpoints and diagnostic logging, and removing the Liger library dependency.
- Sped up image and video requests by **6%** across Instagram, Messenger, Threads, and WhatsApp by rolling out mbedTLS ARMv8 AES acceleration on iOS and Android, validated through microbenchmarks, APK testing, and regression monitoring.
- Unified network security across **15+ Meta apps** by designing, A/B testing, and deploying a centralized HTTP request header security feature spanning Android (Java, Kotlin), iOS (Objective-C), and native (C++) stacks.
- Improved debugging of network and user-experience issues, including **15+ SEV investigations**, by building networking bug-report extensions with diagnostic logging, and owning their rollout across Messenger, Threads, and WhatsApp, with A/B testing for memory regressions.

Amazon May – Aug 2021, 2022, 2023
Software Development Engineering Intern Seattle, WA, USA

- Cut per-ticket handling time by **5 minutes** per developer and sped up infrastructure ticket access for **4 customer teams** by building an automated ticketing system for fault localization and routing on top of my prior-summer monitoring service, using Step Functions, Lambda, and EventBridge Pipes and scoping requirements with **8 stakeholders** through prototypes.
- Enabled system-health analysis across **6 team services** by building the underlying monitoring service, which categorizes ML infrastructure logs through event-driven Step Functions and EventBridge workflows.
- Let additional services integrate with the monitoring service by designing its infrastructure to be modular using SNS and DynamoDB Streams.

EDUCATION

Arizona State University Jan 2023 – Dec 2023
M.S. Computer Science (Thesis) GPA: 4.0

- **Thesis**: *Modeling Spatial Competition and Therapy Protocols in Three-Dimensional Vascularized Cancers* (C++ agent-based simulator).

Arizona State University Aug 2019 – Dec 2022
B.S. Computer Science Summa Cum Laude, GPA: 4.0

- **Achievements**: Moeur Award for 4.0 GPA and New American University Scholar.
- **President, Women in Computer Science**: revived a 100+ member chapter, led hackathons, speaker events, and mentorship programs.

SKILLS

Languages & Frameworks: C, C++, Java, Kotlin, Objective-C, Python, React, TypeScript, JavaScript (D3.js), HTML, CSS, SQL.

Backend & Infrastructure: AWS Lambda, Step Functions, EventBridge, SNS, DynamoDB, REST APIs, Spring Boot, PostgreSQL, distributed systems, CI/CD.

Systems & Reliability: TLS, DNS, transport protocols, mbedTLS, mobile networking, performance profiling, telemetry, A/B testing.

Developer Tools: Git, GitHub, Linux/Unix, Claude, Codex, GitHub Copilot.

PROJECTS

DEVS-Suite Log Visualizer | Java, Spring Boot, PostgreSQL, JavaScript, D3.js

Enabled analysis of timestamped simulation logs through time-based and piecewise-constant charts by building a full-stack application with Spring Boot, PostgreSQL, and D3.js.